PlantDx: Knowledge-Grounded Instruction Tuning for Vision-Language Crop Disease Diagnosis

Aakash Trimbak Jawle
Computer Engineering
Pimpri Chinchwad College of Engineering
Pune, India
aakashjawle101@gmail.com

Kamaaksha Rajendra Kharul
Computer Engineering
Pimpri Chinchwad College of Engineering
Pune, India
kharulkamaaksha@gmail.com

Dr. Santwana Gudadhe
Assistant Professor
Pimpri Chinchwad College of Engineering
Pune, India
santwana.gudadhe@pccoepune.org

***Abstract*- Plant diseases are highly detrimental to the productivity of crops and therefore constitute a threat to food security. Thus, the need for automated disease detection that is not only accurate but also provides sufficient diagnostic information arises. While traditional image-based methods focus on classification and provide only labels without additional diagnostic information, Vision-Language Models can produce more elaborate responses. However, the performance of VLMs in domain-specific tasks like crop disease diagnosis relies on the quality of the fine-tuning supervision data available. In this work, we introduce PlantDx - an open-source pipeline that grounds the adaptation of VLMs to plant diseases from tomato and mango in structured knowledge and instruction-tuning supervision through deterministic generation. We convert a pre-existing Disease Knowledge Base into knowledge representations and an instruction-tuning corpus. In contrast to the previous approaches that use language and vision-language models for generating training captions, our captions do not depend on the input images and are paired with images of the respective crops based on the disease label. The resulting datasets are then employed to QLoRA-fine-tune Qwen2.5-VL-7B-Instruct. Tomato dataset comprises 18,006 images with 10 classes while MangoLeafBD comprises 4,000 images with 8 classes. On the unseen test data, the model attains 93.7% accuracy for tomato (as opposed to 7.3% of the zero-shot one) and 82.0% for mango (as opposed to 17.5%).**

***Keywords*- Vision-Language Models (VLMs), Knowledge-Grounded Instruction Tuning, Disease Knowledge Base, Ontology Construction, Controlled Vocabulary, Multimodal Learning, QLoRA, Qwen2.5-VL, Plant Disease Diagnosis, Agricultural Vision-Language Models.**

I. INTRODUCTION

Some diseases can affect the plants, and these diseases can have an impact on the productivity and profitability of the crops. Tomato and mango come under the list of crops that have several types of diseases and disorders that show similar symptoms and changes with time. It is critical to determine the type of disease in order to manage the crop properly and minimize the risk of losses. However, conventional diagnosis may require visual examination combined with agricultural domain knowledge, which may not always be readily accessible to farmers or agricultural practitioners, particularly in resource-limited settings.

With the integration of deep learning and digital image analysis, plant diseases can be detected automatically from images. Plant disease recognition is a well-established problem in agricultural computer vision and is typically formulated as mapping leaf or plant images to predefined disease categories using classification models. Convolutional Neural Networks (CNNs) and transfer-learning approaches have demonstrated strong classification performance on widely used datasets such as PlantVillage and other crop-specific disease datasets. However, a classification label alone provides limited information about the diagnostic context. The features related to a prediction are not communicated through direct means, while an existing classification model will generate a label and not a structured language output that will serve as a diagnosis to support additional decision-making processes. Also, models created from controlled image sets are not guaranteed to generalize to field conditions where changes in illumination, background, image quality, disease development, and symptoms appearance might influence vision-based recognition.

Unlike conventional closed-set classifiers that are restricted to predefined output labels, VLMs can be adapted to generate natural-language responses conditioned on visual inputs. This capability is relevant to agricultural diagnosis, where a diagnostic

response may need to contain multiple types of information, such as disease identity, observable symptoms, and causal information. However, general-purpose VLMs are not specifically adapted to the visual characteristics, terminology, and disease categories encountered in plant pathology. Consequently, directly applying such models in a zero-shot setting to specialized crop disease diagnosis may result in unreliable predictions, incomplete responses, or inconsistent terminology. The zero-shot evaluation conducted in this study illustrates this limitation: the base Qwen2.5-VL-7B model achieves only 7.3% classification accuracy on the tomato test set and 17.5% on the mango test set.

An important challenge in adapting VLMs to specialized agricultural domains is the construction of reliable instruction-tuning data. Multimodal instruction datasets can be expensive and time-consuming to annotate manually, encouraging the use of automatically generated descriptions as an alternative. However, using another language model or VLM to generate training captions can introduce factual inaccuracies, inconsistent terminology, unsupported statements, or hallucinated disease characteristics into the supervision itself. When such descriptions are subsequently used to fine-tune another model, these errors may be propagated into the resulting model. This creates a need for a training-data construction process in which domain knowledge can be incorporated systematically while maintaining consistency, provenance, validation, and reproducibility.

To address this challenge, this work proposes PlantDx, a knowledge-grounded and reproducible pipeline for adapting Vision-Language Models to structured crop disease diagnosis. Rather than using a language model or VLM to generate training captions, PlantDx begins with a curated and cited Disease Knowledge Base (DKB). The knowledge is deterministically transformed into a typed ontology, controlled vocabulary, symptom lexicon, disease-specific concept models, and a template-based instruction-tuning corpus. The generated corpus is subsequently validated before being used to fine-tune Qwen2.5-VL-7B-Instruct using Quantized Low-Rank Adaptation (QLoRA). By separating agricultural knowledge compilation from model training and inference, PlantDx provides traceability between the underlying domain knowledge and the generated training supervision while allowing the instruction corpus to be regenerated deterministically from the underlying knowledge sources and configuration.

The proposed framework is tested within two crop categories: tomato and mango. There are 18,006 images in the tomato dataset belonging to 10 classes and 4,000 images in the mango dataset belonging to 8 classes. Both crops follow the approach of dataset splitting via deterministically grouped images for constructing the held-out test sets and avoiding any potential image leakage among different splits. Zero-shot learning and fine-tuned models are evaluated based on disease classification metrics and structured text generation metrics for diagnosis. After fine-tuning following knowledge-grounded instruction tuning, the performance of the model reaches 93.7% classification accuracy for tomato and 82.0% for mango, against the respective zero-shot learning accuracies of 7.3% and 17.5%. Evaluation is carried out based on several metrics for the fine-tuned models: BLEU-4, ROUGE-L, METEOR, CIDEr, and BERTScore-F1.

The key contributions of this research include:

1. A knowledge-grounded instruction-tuning pipeline for crop disease diagnosis: PlantDx transforms a curated agricultural Disease Knowledge Base into structured multimodal training supervision without using an LLM or VLM in the caption-generation path, providing traceability between the underlying domain knowledge and the generated supervision.

2. A deterministic and reproducible data-generation framework: The proposed framework integrates ontology construction, controlled vocabulary and symptom lexicons, disease-specific concept modeling, template-based generation, and validation to produce consistent instruction-tuning data from structured agricultural knowledge.

3. Parameter-efficient adaptation of a general-purpose VLM: Qwen2.5-VL-7B-Instruct is adapted for crop disease diagnosis using QLoRA, providing a parameter-efficient mechanism for domain specialization without requiring full-model parameter updates.

4. Evaluation across two crop domains: The framework is empirically evaluated on tomato and mango disease datasets, demonstrating the applicability of the same knowledge-grounded adaptation pipeline across two different crop domains.

5. Joint evaluation of classification and diagnostic text generation: In addition to disease classification accuracy and Macro-F1, the study evaluates generated diagnostic descriptions using multiple text-generation metrics, providing complementary measurements of disease recognition and generated response quality.

6. Traceable and reproducible agricultural supervision: By deriving training examples from structured disease knowledge rather than model-generated captions, PlantDx provides a deterministic mechanism for constructing domain-specific VLM supervision and a foundation for extending the methodology to additional crop and disease domains.

The remainder of this paper is organized as follows. Section 2 reviews related work in plant disease

classification, Vision-Language Models, knowledge-grounded learning, and parameter-efficient fine-tuning. Section 3 describes the PlantDx methodology, including the Disease Knowledge Base, ontology, controlled vocabulary and symptom lexicon, disease-specific concept modeling, caption generation and validation, instruction-tuning dataset construction, and model adaptation. Section 4 presents the datasets, experimental configuration, training procedure, and evaluation methodology. The experimental findings of tomatoes and mangoes are reported in section 5. Section 6 is devoted to the discussion of results, implications, limitations and possible sources of errors. Section 7 provides the conclusion of the paper and suggests future research direction.

The complete framework consists of two interconnected workflows: a knowledge-to-dataset workflow and a model adaptation and evaluation workflow. In the first workflow, crop-specific disease knowledge is organized within a structured Disease Knowledge Base and transformed into a typed ontology, controlled vocabulary and symptom lexicon, and disease-specific concept models. These representations are passed to a template-based generation process that produces structured diagnostic captions. The generated captions are subsequently validated to construct the final instruction-tuning corpus. Importantly, this process does not invoke an LLM or VLM and does not access image pixels during caption generation. Consequently, the textual supervision is derived from explicit agricultural knowledge rather than from model-generated interpretations of images.

In the second workflow, the generated instruction corpus is paired with the corresponding crop images to construct the multimodal training datasets. The resulting datasets are used to parameter-efficiently fine-tune Qwen2.5-VL-7B-Instruct using Quantized Low-Rank Adaptation (QLoRA). The adapted models are evaluated against the original zero-shot foundation model using held-out test sets for both tomato and mango. Evaluation considers disease classification performance as well as the quality of the generated structured diagnostic descriptions.

The overall PlantDx workflow can be summarized as:

Disease Knowledge Base -> Ontology -> Controlled Vocabulary -> Concept Models -> Template-Based Generation -> Validated Caption Corpus -> QLoRA Fine-Tuning -> Evaluation

Though the general pipeline remains the same for both crops, the specifics regarding disease, classification, vocabulary, and instruction are dependent on the crop. This enables the framework to accommodate different disease taxonomies while maintaining a common methodology for knowledge

II. METHODOLOGY

A. PlantDx Overview

PlantDx is a knowledge-grounded Vision-Language Model (VLM) pipeline designed for structured crop disease diagnosis across tomato and mango. The framework is based on the principle that the quality, consistency, and provenance of the supervision used for model adaptation are important to the effectiveness of the adapted VLM. Instead of generating training captions using another language model or vision-language model, PlantDx derives its instruction-tuning supervision from a curated and referenced Disease Knowledge Base through a deterministic data-generation process.

compilation, supervision generation, model adaptation, and evaluation.

The framework is implemented as a modular pipeline in which the major stages of knowledge processing, dataset construction, model adaptation, and evaluation are separated. The deterministic generation process allows the instruction corpus to be reconstructed from the underlying structured knowledge and generation configuration. This separation provides traceability between agricultural knowledge and training supervision while enabling the same methodology to be applied consistently across the tomato and mango domains.

B. PlantDx Architecture

The overall architecture of PlantDx is shown in Figure. 1. The pipeline consists of two parallel processing paths: a knowledge-processing path and a dataset-processing path. These paths are combined during instruction-tuning dataset construction before model fine-tuning.

The knowledge-processing path begins with the Disease Knowledge Base, which contains the curated disease information used by the pipeline. This knowledge is passed through the Ontology Compiler to obtain a structured representation of the disease domain. The resulting information is then used to construct a controlled vocabulary and symptom lexicon, followed by disease-specific concept models. These components provide the structured information required by the subsequent template-based generation process.

The Template Engine uses the concept models and controlled terminology to generate candidate diagnostic captions. These candidates are processed by the Caption Generator and Validator, producing the final Caption Corpus. This process is designed to generate the instruction-tuning text from the structured disease knowledge rather than from model-generated image descriptions.

In parallel, the raw PlantVillage tomato dataset and MangoLeafBD dataset pass through an Audit stage followed by Normalization. The audit stage is used to inspect the input datasets, while normalization prepares the image data in a consistent format for subsequent processing.

The normalized images and validated caption corpus are combined by the Dataset Builder, which constructs the final image-caption instruction-tuning dataset. This dataset is then used for QLoRA fine-tuning of Qwen2.5-VL on MLX. The resulting fine-tuned model is evaluated against the base model using the evaluation pipeline. Finally, the trained adapters are used in the Streamlit demonstration, providing an interactive interface for crop disease diagnosis.

Thus, the architecture separates knowledge compilation, image-data preparation, dataset construction, model adaptation, and evaluation into distinct stages. This separation allows the instruction-tuning corpus to be generated through a reproducible pipeline before it is combined with the image data for model training.

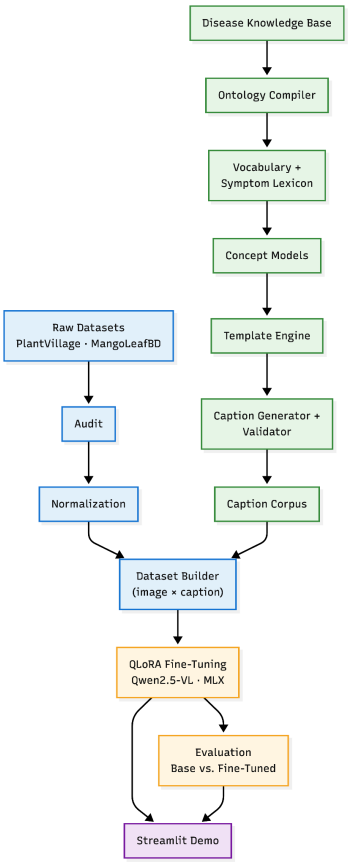


Fig. 1. Overall architecture of the PlantDx knowledge-grounded vision-language model pipeline

C. Disease Knowledge Base

The Disease Knowledge Base (DKB) is the source of agricultural knowledge that is used in the whole process of the PlantDx pipeline. The DKB has disease records of crop-specific in the domains of tomato and mango and forms the factual foundation for building the ontology, controlled vocabulary, symptom lexicon, concept model of disease-specific and captions for instruction tuning. The DKB is curated from referenced agricultural sources so that the information used to construct the training supervision remains traceable to its underlying evidence.

Each disease record represents a structured description of a crop disease or condition and contains information related to disease identity, terminology, causal information, characteristic symptoms, observable features, controlled vocabulary, and related knowledge constraints. The structured representation allows the same underlying disease information to be consumed consistently by the downstream ontology and caption-generation stages

rather than being independently rewritten for each stage.

The DKB also distinguishes information that is appropriate for the target single-leaf image setting from information that cannot be established from the image type. This distinction is important because some diseases may have symptoms affecting other plant parts or broader plant-level effects. Such information is not automatically treated as an observable property of an individual leaf. The DKB therefore provides the basis for restricting generated supervision to concepts appropriate for the intended diagnostic setting.

In addition to positive disease information, the knowledge representation includes controlled terminology and constraints on inappropriate or unsupported terms. These constraints are subsequently used by the vocabulary, concept-model, and validation stages to prevent unsupported disease descriptions from entering the final caption corpus.

The DKB functions as a single source of truth for the knowledge-to-supervision pipeline. The caption-generation process does not obtain disease information from an LLM or VLM and does not infer disease characteristics from image pixels. Instead, disease-specific information is selected from the structured knowledge representation and passed through the downstream compilation and generation stages. Consequently, the generated captions can be associated with the corresponding disease knowledge and its cited sources.

For the experiments reported in this study, the DKB covers 10 tomato classes and 8 mango classes, including healthy conditions and disease categories. The same DKB structure is used across both crop domains, while the disease records, terminology, and associated symptoms remain crop-specific. This allows PlantDx to maintain a common knowledge-compilation procedure while accommodating the distinct disease taxonomies of tomato and mango.

D. Ontology and Controlled Vocabulary

The Disease Knowledge Base with its structure is put together in a typed ontology which illustrates the relationship between the crops, diseases, symptoms, observable features, and other terms related to diseases. It acts as the representation level between the knowledge records and the caption generation process.

The ontology preserves crop-specific disease concepts while following a common representation scheme across tomato and mango. Disease-related concepts are organized according to their semantic roles, and relationships between concepts are explicitly represented. This structured representation allows downstream components to identify which concepts are associated with a particular disease and which concepts are permitted or restricted in the target diagnostic setting.

A controlled vocabulary and symptom lexicon are derived from the structured knowledge representation. The controlled vocabulary constrains the terminology available to the caption-generation process, while the symptom lexicon provides bounded lexical realizations for disease-related visual characteristics. This reduces the dependence on unrestricted natural-language generation and helps maintain consistent terminology across captions.

The vocabulary also incorporates disease-specific and domain-specific constraints, including permitted terminology and forbidden terms. These constraints are used during caption construction and subsequent validation to prevent incompatible or unsupported terminology from entering the final instruction-tuning corpus. The resulting knowledge representation therefore provides both the concepts required to describe a disease and the lexical constraints governing how those concepts may be expressed.

The ontology, controlled vocabulary, and symptom lexicon are generated deterministically from the Disease Knowledge Base. They therefore remain linked to the underlying agricultural knowledge rather than being independently authored during caption generation. This intermediate structured representation forms the basis for the disease-specific concept modeling and template-based caption generation described in the following section.

E. Disease Concept Models and Caption Generation

The structured disease representations are further converted into disease-specific concept models that define the information permitted in the generated diagnostic captions. For each disease, the concept model distinguishes between mandatory, optional, and forbidden concepts. Mandatory concepts represent information that should be retained in the diagnostic description, optional concepts provide controlled variation between captions, and forbidden concepts prevent incompatible or unsupported information from being included.

The concept models are used together with a template-based generation framework to construct the instruction-tuning captions. The template library contains 33 templates organized into eight template families, providing controlled variation in sentence structure and descriptive style. Templates define the linguistic structure of a caption, while the disease-specific concept models determine the disease-related information that can populate the corresponding fields. This separation allows linguistic variation to be introduced without allowing

unrestricted disease knowledge to enter the generation process.

Caption generation is performed without accessing image pixels and without invoking an LLM or VLM. Instead, a disease label is used to select the corresponding concept model and knowledge-grounded concepts from the compiled representations. These concepts are then realized through compatible templates and controlled vocabulary. Consequently, the generated captions describe information licensed by the disease class and the underlying Disease Knowledge Base rather than attempting to infer image-specific observations during caption construction.

The generation process also incorporates controlled variation in the selected concepts and their textual realization. This allows multiple captions to be produced for the same disease while maintaining consistency with the underlying knowledge constraints. The resulting captions therefore provide structured natural-language supervision while retaining a direct relationship with the disease knowledge from which they were generated.

Before inclusion in the final instruction-tuning corpus, generated captions are subjected to the validation process described in the following section. Captions that violate the defined conceptual, terminology, observability, or structural constraints are excluded from the validated corpus.

F. Caption Validation, Provenance, and Reproducibility

The generated captions are subjected to a dedicated validation stage before being included in the instruction-tuning corpus. The validation process checks whether each caption conforms to the disease-specific concepts and constraints established during knowledge compilation and caption generation. The validation framework consists of 12 checks covering ontology legality, template legality, mandatory concept presence, forbidden concept absence, forbidden-term usage, duplicate wording, grammatical validity, modifier legality, observability, confidence consistency, severity constraints, and traceability.

The validation process is designed to ensure that generated captions remain consistent with the structured agricultural knowledge from which they were derived. In particular, the observability constraint restricts captions to information appropriate for the target single-leaf image setting, while the severity constraint prevents unsupported per-image severity claims when such annotations are not available in the source datasets. Terminological constraints further prevent disease descriptions from using incompatible or unsupported vocabulary.

Captions that fail the validation requirements are excluded from the final corpus. The accepted captions retain their association with the underlying disease concepts and knowledge sources, providing provenance for the generated supervision. The resulting corpus is therefore not treated as an unrestricted collection of generated text, but as a validated knowledge-derived dataset.

The caption-generation and validation pipeline is deterministic and uses fixed configuration and seeded generation. The intermediate knowledge artifacts and generated corpus are content-hashed, allowing the resulting supervision to be reconstructed from the underlying knowledge and configuration. This provides reproducibility across dataset-generation runs and reduces dependence on manually edited training descriptions.

For the current experiments, the validated corpus contains 1,966 accepted captions covering the tomato and mango disease classes. The final corpus is subsequently used as a frozen source of textual supervision during multimodal instruction-dataset construction.

G. Instruction-Tuning Dataset Construction

The validated caption corpus is combined with the normalized tomato and mango image datasets to construct the multimodal instruction-tuning datasets. For each image, the dataset builder uses the corresponding disease label to associate the image with an appropriate caption from the frozen corpus. The caption is used as the target response, while a crop-specific diagnostic instruction is provided as the input prompt.

Each training instance therefore consists of an image, an instruction, and a target diagnostic response. The target response is taken directly from the validated caption corpus rather than being regenerated during dataset construction. The image contributes the visual input to the VLM, while the disease label determines the corresponding knowledge-grounded diagnostic target. Importantly, the caption itself is generated independently of the image pixels; the image-caption association is established during construction of the multimodal training dataset.

The dataset construction process is deterministic and uses the normalized image paths, disease labels, validated caption corpus, and predefined instruction set. The resulting records are divided into training, validation, and held-out test partitions using the predefined deterministic splitting procedure. The same overall construction methodology is applied to both crops while using their respective disease classes, images, and crop-specific instructions.

For the experiments performed, the tomato database contains a total of 18,006 images, where 16,201 are

for the training database, 895 images for the validation database, and 910 images for the testing database. For the mango database, there are a total of 4,000 images, where 3,600 are for the training database and 200 images each for the other databases. The resulting multimodal datasets provide the supervision used for QLoRA fine-tuning of Qwen2.5-VL-7B-Instruct.

H. QLoRA Fine-Tuning

The constructed instruction-tuning datasets are used to adapt Qwen2.5-VL-7B-Instruct-4bit for crop disease diagnosis using Quantized Low-Rank Adaptation (QLoRA). QLoRA enables parameter-efficient adaptation by keeping the quantized base model fixed while learning a comparatively small set of low-rank update parameters. Adapters are independently trained on the tomato and mango datasets, enabling the use of the same architecture of the VLM with its training on crop-specific instruction tuning datasets.

The settings for the LoRA are: rank = 16, alpha = 32, and dropout = 0.05. Adaptation is performed only for the language model projection layers – query, key, value, and output projections, while the vision part of the model stays unchanged (is frozen). The resulting adapters therefore modify the language-side parameters responsible for generating the structured diagnostic responses without updating the complete set of base-model parameters.

Training is done through the MLX training pipeline in the context of an Apple Silicon machine with 24GB unified memory. The optimizer configuration involves the learning rate set to 1×10^{-4} and cosine learning rate scheduler with 100 warm up steps. Batch size during training is 1 with 8 steps of gradient accumulation giving us batch size of 8 and gradient clip at 1.0. Max sequence length is set to 2048 tokens with the input image resolution being 448 x 448.

Following training, the resulting crop-specific adapters are evaluated against the corresponding zero-shot base model. The base and fine-tuned models are evaluated using the same held-out test data and diagnostic instruction format, enabling the effect of knowledge-grounded instruction tuning to be assessed under a consistent evaluation setting.

I. Evaluation Methodology

The performance of the proposed PlantDx models is evaluated by comparing the zero-shot Qwen2.5-VL-7B-Instruct-4bit model with the corresponding QLoRA-fine-tuned model for tomato and mango. Both models are evaluated on the same held-out test sets using the same diagnostic instruction format, allowing the effect of domain-specific instruction tuning to be assessed under a consistent evaluation setting. Decoding is performed deterministically without sampling.

1) Classification Evaluation

Disease recognition performance is evaluated using accuracy, Macro-F1, and Weighted-F1, together with per-class performance analysis. Accuracy measures the proportion of correctly identified disease classes across the test set. Macro-F1 is calculated as the unweighted mean of the F1 scores across classes, giving equal importance to each disease category, while Weighted-F1 accounts for the number of samples belonging to each class.

Per-class results and confusion matrices are also generated to identify classes for which the fine-tuned model performs particularly well or remains challenging. This provides a more detailed view of disease recognition than overall accuracy alone.

2) Diagnostic Caption Evaluation

Since PlantDx is designed to produce structured diagnostic text and not just disease labels, the output is further evaluated on the basis of the captions contained in the validated dataset. Caption quality is measured by BLEU-4, ROUGE-L, METEOR, CIDEr, and BERTScore-F1. These metrics provide complementary measures of lexical overlap, sequence similarity, semantic similarity, and correspondence between generated and reference diagnostic descriptions.

The caption evaluation is performed for both the zero-shot and fine-tuned models, allowing changes in the quality of structured diagnostic responses to be measured alongside changes in disease classification performance.

3) Knowledge and Output Consistency Evaluation

In addition to standard classification and caption-generation metrics, the evaluation pipeline performs deterministic checks of generated responses against the knowledge-derived terminology and constraints. These checks examine potential issues such as incorrect disease mentions, unsupported symptoms, incorrect causal information, inappropriate non-leaf observations, and other knowledge or terminology inconsistencies.

These checks are based on the frozen Disease Knowledge Base and its compiled vocabulary artifacts rather than an LLM-based judge. They are therefore used to assess whether generated responses remain consistent with the structured agricultural knowledge used to construct the training supervision.

4) Statistical and Performance Analysis

The evaluation pipeline also produces per-disease metric breakdowns and statistical comparisons between the zero-shot and fine-tuned outputs. In

addition to diagnostic quality, inference performance is recorded using measures such as generation latency and throughput-related statistics. These measurements provide a broader assessment of the effect of QLoRA adaptation beyond classification accuracy and caption quality.

TABLE II

DATASET DISTRIBUTION AND EXPERIMENTAL SPLITS

Dataset	Classes	Total Images	Train	Validation	Test
Tomato	10	18006	16201	895	910
Mango	8	4000	3600	200	200

This is a summary of the number of classes and the total number of images that were employed in the experiments conducted on tomatoes and mangoes, together with the deterministic training, validation, and test sets. The dataset of tomato consists of 18,006 images distributed over 10 classes, whereas that of mango consists of 4,000 images over 8 classes.

TABLE III

EXPERIMENTAL CONFIGURATION

Parameter	Value
Model	Qwen2.5-VL-7B-Instruct-4bit
Fine-tuning	QLoRA
LoRA (r / α / dropout)	16 / 32 / 0.05
Target modules	q_proj, k_proj, v_proj, o_proj
Vision encoder	Frozen
Learning rate	$1 \times 10^{-4} \rightarrow 1 \times 10^{-6}$
Scheduler	Cosine
Warm-up	100 steps
Batch / effective batch	1 / 8
Gradient accumulation	8
Gradient clipping	1
Epochs	1
Max sequence length	2048
Image size	448 × 448
Seed	20260711
Framework	MLX
Hardware	Apple Silicon M4 Pro
Memory	24 GB

The table summarizes the configurations used to adapt Qwen2.5-VL-7B-Instruct for tomato and mango disease diagnosis using crop-specific QLoRA adapters.

III. EXPERIMENTAL SETUP

A. Datasets

To evaluate the efficacy of our PlantDx approach across multiple crops, two crop types have been selected – tomato and mango. In this section, two datasets are utilized, the tomato subset of the PlantVillage dataset and the MangoLeafBD dataset. Both datasets contain the leaf images for the predefined disease classes and healthy state of each crop and used to generate the crop-specific instruction-tuning datasets, as discussed in Section 3.

1) Tomato Dataset

In tomato studies, we apply tomato sub-dataset from PlantVillage that contains 18,006 images within 10 classes: healthy tomato, bacterial spot, early blight, late blight, leaf mold, septoria leaf spots, spider mites, target spot, mosaic virus, and yellow leaf curl virus.

The process of train/validation/test sets creation is done using the deterministic image-grouped splitting process as mentioned in Section 3. As a result, we obtain the train set having 16,201 images, the validation set having 895 images, and the test set having 910 images. Our test set is isolated from the other datasets and will be used at the end of our evaluation of two models.

2) Mango Dataset

For mango classification experiments, we use the MangoLeafBD dataset having 4,000 samples belonging to 8 classes – healthy, anthracnose, bacterial canker, cutting weevil, dieback, gall midge, powdery mildew, and sooty mould.

Through the process of deterministic split, we divide the dataset into 3,600 training samples, 200 validation samples, and 200 testing samples. This testing sample will be used for evaluation of the zero-shot base model and mango-adapted QLoRA model.

3) Dataset Split

For both crop domains, the dataset construction process involves the 90:5:5 training-validation-test split. The assignment of images is deterministic and image-grouped to avoid image leakage across different partitions. The same test sets are kept out for both cases to measure performance of both models.

B. Experimental Configuration

The experiments use Qwen2.5-VL-7B-Instruct-4bit as the base Vision-Language Model and are performed using the MLX framework on an Apple Silicon system with 24 GB of unified memory. The model is adapted separately for the tomato and mango domains using QLoRA, while the base model remains quantized and the vision component is kept frozen. The same overall fine-tuning configuration is used across the two crop domains.

The principal model, optimization, and runtime parameters used in the experiments are summarized in Table 2.

C. Evaluation Metrics

The performance of PlantDx is evaluated from two perspectives: disease classification and generation of structured diagnostic text. The comparison is made between the zero-shot base model and the QLoRA-fine-tuned one using the same test datasets for tomato and mango.

1) Classification Metrics

For measuring the classification, Accuracy, Macro-F1 and Weighted-F1 will be considered. The accuracy refers to the number of sample points that are correctly classified. The Macro-F1 score is computed as the unweighted mean of F1 scores of all disease classes, meaning that all disease classes have the same weight. The Weighted-F1 metric is computed as the mean F1 score, where the weights are determined by the number of samples per each disease class.

The classification metrics are additionally supplemented by per-class performance analysis and confusion matrices, which help to find out which disease classes are still hard to differentiate even after fine-tuning.

2) Caption-Quality Metrics

Since PlantDx is capable of generating structured diagnostic responses other than assigning diseases to these responses, these responses are assessed by the following evaluation criteria BLEU-4, ROUGE-L, METEOR, CIDEr and BERTScore-F1 comparing generated response to the reference caption.

BLEU-4 evaluates n-gram overlap between the generated and reference captions. ROUGE-L evaluates the similarity using the longest common subsequence of sequences. METEOR evaluates the unigram matching; however, it adds additional weights to that metric. CIDEr evaluates the similarity between the generated description and reference descriptions through the use of TF-IDF weighted n-grams. BERTScore-F1 evaluates the semantic similarity of the captions by utilizing the contextual embeddings.

3) Knowledge Consistency Evaluation

The evaluation pipeline also performs deterministic consistency checks of the generated responses compared to the Disease Knowledge Base and vocabulary artifacts that were built. The checks are aimed at detecting inconsistency issues like disease mentions errors, unsupplied symptoms, wrong causality information, inapplicable non-leaf observations, etc.

The checks are not based on the judge from the large language model. Instead, the fixed knowledge and vocabulary artifacts of the PlantDx pipeline are used to determine whether the generated responses are still consistent with the knowledge used to create the instruction-tuning supervision.

4) Statistical Analysis

The evaluation pipeline also performs the per-disease analysis and statistical comparison between the

zero-shot and fine-tuned models. The analysis is aimed at finding out whether the observed differences between the generated text metrics are statistically meaningful.

D. Baseline and Comparison Protocol

To evaluate the effect of knowledge-grounded instruction tuning, the fine-tuned models are compared with the corresponding zero-shot Qwen2.5-VL-7B-Instruct-4bit base model. The base model is evaluated without crop-specific fine-tuning, while separate QLoRA adapters are evaluated for the tomato and mango domains.

In both cases, the test sets and instructions remain the same for the testing procedure. In other words, all differences in performance are measured on equal input grounds. For both models, decoding is performed deterministically without sampling.

For each test image, the model receives the image together with the corresponding diagnostic instruction and generates a structured response. The generated responses are then processed using the same evaluation pipeline to obtain classification metrics, caption-quality metrics, and deterministic knowledge-consistency measures.

TABLE IV

EVALUATION METRICS

Category	Metric	Purpose
Classification	Accuracy	Overall proportion of correctly classified samples
Classification	Macro-F1	Equal-weighted F1 across disease classes
Classification	Weighted-F1	F1 weighted by class support
Text Generation	BLEU-4	N-gram overlap between generated and reference captions
Text Generation	ROUGE-L	Longest common subsequence similarity
Text Generation	METEOR	Token-level similarity with linguistic matching
Text Generation	CIDEr	TF-IDF-weighted similarity to reference descriptions
Text Generation	BERTScore-F1	Semantic similarity between generated and reference text
Knowledge Consistency	Deterministic consistency checks	Detection of knowledge, terminology, and observability violations

The table summarizes metrics for disease recognition, structured diagnostic text generation, and knowledge consistency with PlantDx's agricultural knowledge base.

TABLE V

BASELINE AND COMPARISON PROTOCOL

Component	Zero-shot Baseline	Fine-tuned Model
Base model	Qwen2.5-VL-7B-Instruct-4bit	Qwen2.5-VL-7B-Instruct-4bit
Crop-specific fine-tuning	None	QLoRA
Adapter	None	Crop-specific LoRA adapter
Test data	Held-out test set	Same held-out test set
Instruction format	Same	Same
Decoding	Deterministic, no sampling	Deterministic, no sampling
Evaluation metrics	Same	Same

The table compares the zero-shot and QLoRA-fine-tuned models under identical test and decoding conditions.

The comparison therefore follows the following evaluation setting:

Zero-shot Qwen2.5-VL-7B-Instruct -> Held-out Test Set versus QLoRA Fine-tuned Qwen2.5-VL-7B-Instruct -> Same Held-out Test Set

The comparison is performed independently for tomato and mango, with each fine-tuned model evaluated using the corresponding crop-specific test set.

IV. RESULTS

A. Overall Classification Results

Classification performance of the zero-shot and QLoRA fine-tuned Qwen2.5-VL-7B-Instruct models was measured on the held-out test sets for both tomato and mango crops. Overall classification performance metrics in terms of accuracy, Macro-F1, and Weighted-F1 are summarized in Table 5.

For the tomato test set, the zero-shot model had accuracy of 7.3%, Macro-F1 of 0.034 and Weighted-F1 of 0.025. Fine-tuning of the model using the PlantDx instruction-tuning dataset led to obtaining the accuracy of 93.7% for the tomato test set. In addition, Macro-F1 and Weighted-Diff. were enhanced to 0.919 and 0.837, respectively.

TABLE VI

OVERALL CLASSIFICATION PERFORMANCE OF ZERO-SHOT AND QLORA FINE-TUNED MODELS

Dataset	Model	Accuracy	Macro-F1	Weighted-F1
Tomato	Zero-shot	7.30%	0.034	0.025
Tomato	QLoRA fine-tuned	93.70%	0.919	0.937
Mango	Zero-shot	17.50%	0.128	0.128
Mango	QLoRA fine-tuned	82.00%	0.811	0.811

Overall disease-classification performance of the zero-shot Qwen2.5-VL-7B-Instruct model and the corresponding crop-specific QLoRA fine-tuned models on the held-out tomato and mango test sets.

While Macro-F1 and Weighted- F1 were only 0.128 for each type, the zero-shot model was more accurate at 17.5% in the mango test set. Fine-tuning resulted in an accuracy of 82.0%, while Macro-F1 and Weighted-Different-Tilting were both scored as high at 0.811.

It should be noted that the fine-tuned models did better than the test sets using the zero-shot model. The accuracy of the tomato model was improved by 86.4%, whereas the mango model had an even greater improvement of 64.5%.

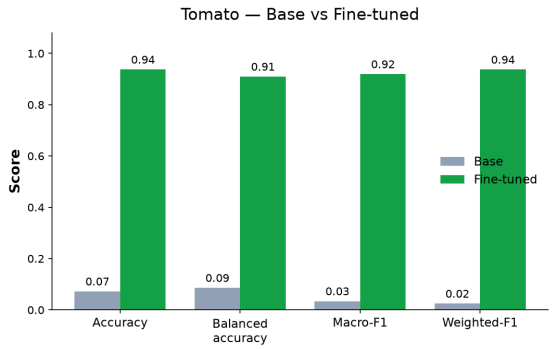


Fig. 2. Tomato: Zero-Shot vs. QLoRA Fine-Tuned Performance

Fig. 2. Comparison of zero-shot and QLoRA fine-tuned Qwen2.5-VL-7B-Instruct performance on the tomato held-out test set. The fine-tuned model shows substantial improvement across accuracy, Macro-F1, and Weighted-F1.

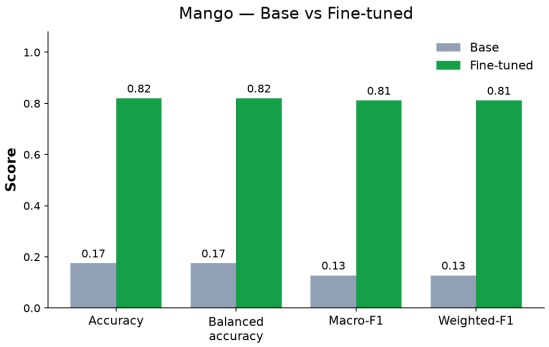


Fig. 3. Mango: Zero-Shot vs. QLoRA Fine-Tuned Performance

Fig. 3. Comparison of zero-shot and QLoRA fine-tuned Qwen2.5-VL-7B-Instruct performance on the mango held-out test set. The fine-tuned model substantially outperforms the zero-shot baseline across the reported classification metrics.

B. Per-Class Classification Results

Overall classification metrics provide a summary of model performance, but they do not show how the model performs for individual disease categories. Therefore, the per-class F1 scores and confusion matrices of the QLoRA fine-tuned models were examined separately for the tomato and mango test sets.

1) Tomato

The finely-tuned tomato model performed exceptionally well in both disease and healthy groups.. Yellow Leaf Curl Virus scored highest in terms of F1 with an overall score of 0.994, followed by Mosaic (0.974), Healthy (0.83), and Bacterial Spot. The F1 scores for Septoria Leaf Spot, Leaf Mold, Late Blight, and Spider Mites were 0.924, 0.925, 0.821, 2.06, and 0.717, respectively. Target Spot and Early Blight had lower F1 scores, with the former scoring 0.845 and the latter scoring 0.76.

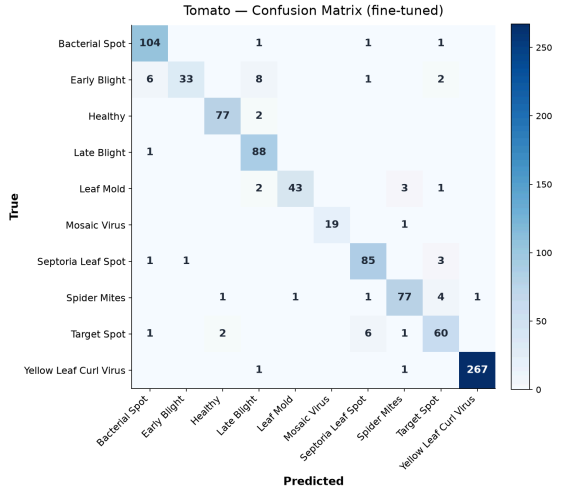


Fig. 4. Tomato confusion matrix of the fine-tuned model

Fig. 4. Confusion matrix of the QLoRA fine-tuned Qwen2.5-VL-7B-Instruct model on the tomato held-out test set. Rows represent the true disease classes and columns represent the predicted classes.

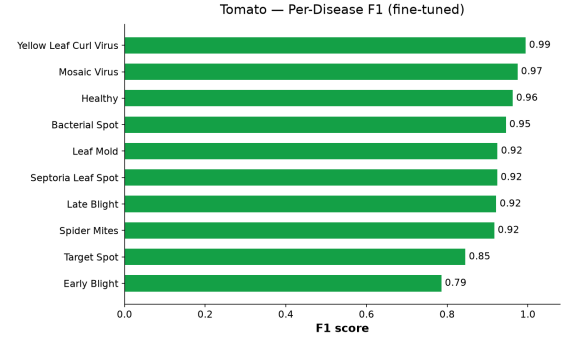


Fig. 5. Per-class F1 scores for tomato

Fig. Per-class F1 scores of the QLoRA fine-tuned Qwen2.5-VL-7B-Instruct model for the tomato held-out test set. The figure shows the variation in classification performance across the 10 tomato disease and healthy classes.

2) Mango

The mango results show greater variation in performance across the eight classes. Healthy and Cutting Weevil had the best F1 score at 1.000. On the other hand, the classes of Bacterial Canker and Powdery Mildew both scored F1 of 0.958, while Sooty Mould and Gall Midge got 0.926 and 0.862, respectively.

The class with the least accuracy was Anthracnose, whose F1 score was 0.708. The class with the least performance was Die Back with an F1 score of 0.077. The confusion matrix of the predicted classes is provided in Figure 5, while Figure 6 shows the F1 scores for each mango class.

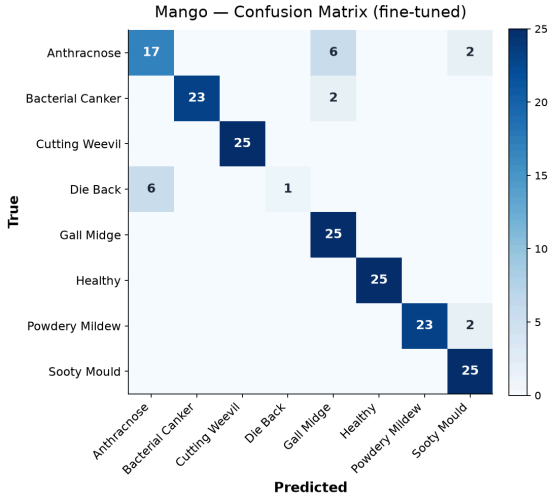


Fig. 6. Mango confusion matrix of the fine-tuned model.

Fig. 6. Confusion matrix of the QLoRA fine-tuned Qwen2.5-VL-7B-Instruct model on the mango held-out test set. Rows represent the true disease classes and columns represent the predicted classes.

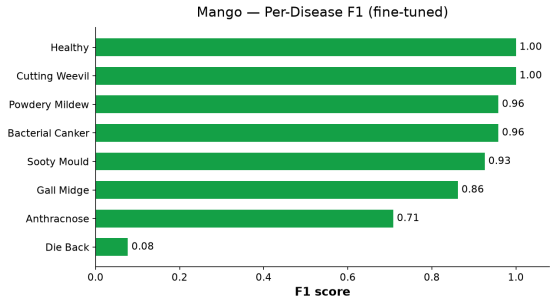


Fig. 7. Per-class F1 scores for mango

Fig. 7. Per-class F1 scores of the QLoRA fine-tuned Qwen2.5-VL-7B-Instruct model for the mango held-out test set. The figure shows the variation in classification performance across the eight mango disease and healthy classes.

The per-class results show that the fine-tuned model does not perform uniformly across all disease categories. While several classes achieve F1 scores above 0.90, lower scores are observed for Early Blight and Target Spot in tomato and particularly for Anthracnose and Die Back in mango. These class-level differences are considered further in the Discussion section.

C. Caption Quality for Diagnostics

The models' ability to accurately reproduce the anticipated diagnostic information was evaluated alongside the classification performance of the generated diagnostic captions. In this analysis, five metrics of caption quality were used: BLEU-4, ROUGE-L, METEOR, CIDEr, and BERTScore-F1. The results obtained for the zero-shot and fine-tuned models using QLoRA are presented in Table 6.

The fine-tuned model is more effective than the zero-shot model for tomato on all five metrics. All five caption-quality metrics for the tomato dataset were superior to those of the fine-tuned model, with a significant improvement over the zero-shot model. From a low point of 0.93, BLEU-4 was higher than 0.093 and ROUGE-L was lower than 0,192. The fine-tuned model was more effective than the zero-shot model in terms of caption-quality metrics for both datasets. Improvements were seen in the following structures: BLEU-4, ROUGE-L (both contain hydrogen bonds), METEOR, CIDEr, and BERTScore-F1. The comprehensive conclusions are presented in Table 6.

Tomato and mango exhibited a similar pattern. Ultimately, adjustment resulted in an improvement in the similarity of generated diagnostic responses with reference captions.

The scores for the fine-tuned models are higher than those for their zero-shot models across all five caption-quality measures for both crops.

TABLE VII

CAPTION-QUALITY COMPARISON BETWEEN ZERO-SHOT AND QLORA FINE-TUNED MODELS FOR TOMATO AND MANGO

Dataset	Model	BLEU-4	ROUGE-L	METEOR	CIDEr	BERTScore-F1
Tomato	Zero-shot	0.003	0.093	0.149	0	0.843
Tomato	QLoRA fine-tuned	0.192	0.428	0.431	0.956	0.907
Mango	Zero-shot	0.004	0.098	0.16	0	0.843
Mango	QLoRA fine-tuned	0.19	0.423	0.454	0.946	0.907

The table compares the caption-generation performance of the zero-shot and QLoRA fine-tuned models across tomato and mango using BLEU-4, ROUGE-L, METEOR, CIDEr, and BERTScore-F1

V. DISCUSSION

The results show a clear improvement after QLoRA fine-tuning for both crops. The zero-shot Qwen2.5-VL-7B-Instruct model achieved an accuracy of 7.3% for tomato and 17.5% for mango. After fine-tuning with the PlantDx instruction-tuning datasets, the accuracy increased to 93.7% for tomato and 82.0% for mango. The increase in Macro-F1 and Weighted-F1 also shows that the improvement was reflected beyond overall accuracy.

The same PlantDx pipeline was used for both crops. The disease information and terminology were adapted to the respective crop, while the overall knowledge-compilation and QLoRA fine-tuning process remained the same. The results on both datasets therefore show that the pipeline can be applied to different crop disease domains without changing its overall design.

The tomato and mango results are not identical, however. The tomato model reached a higher final accuracy of 93.7%, whereas the mango model reached 82.0%. The per-class results help explain this difference. Most tomato classes achieved high F1 scores, although some classes were more difficult to classify than others. In the mango results, the performance varied more strongly between classes, with Die Back showing particularly low performance. This suggests that some disease categories remain difficult for the model even after fine-tuning.

Caption generation also shows a corresponding improvement. In the case of tomato and mango, the fine-tuned models scored better than the zero-shot models for the evaluation metrics like BLEU-4, ROUGE-L, METEOR, CIDEr, and BERTScore-F1. Since PlantDx is intended to produce structured diagnostic responses rather than only disease labels, these results provide an additional view of the effect of fine-tuning. The higher scores indicate better

agreement between the generated responses and the reference captions used during evaluation.

A key part of the PlantDx approach is the way the training captions are created. They are not generated by another VLM or language model. Instead, they are produced from the curated Disease Knowledge Base through the ontology, controlled vocabulary, disease-specific concept models, templates, and validation stages. This gives the generated supervision a direct connection to the underlying disease knowledge. However, the current experiments evaluate the complete PlantDx pipeline and do not include ablation studies. Therefore, the results cannot be used to determine the contribution of each individual component separately.

The evaluation results should also be considered in the context of the datasets used. The models were evaluated on held-out test images from the respective datasets, with the splitting procedure designed to avoid leakage between the partitions. However, these experiments do not measure performance on uncontrolled field images or images captured under substantially different conditions. Therefore, the 93.7% tomato accuracy, for example, should not be interpreted as an equivalent expected accuracy for real-world agricultural images.

Overall, the results indicate that the combination of structured agricultural knowledge, deterministic instruction and data construction, and VLM adaptation based on QLoRA was beneficial for both disease classification and diagnostic text generation for the tested tomato and mango datasets. However, differences between disease classes indicate that there are still difficulties to overcome.

VI. CONCLUSION

This study presented PlantDx, a knowledge-grounded and reproducible pipeline for adapting Vision-Language Models to crop disease diagnosis. Instead of relying on another language model or VLM to create the training captions, PlantDx uses a curated and cited Disease Knowledge Base and converts it into structured knowledge, controlled terminology, disease-specific concepts, and validated captions through a deterministic process.

The resulting instruction-tuning data was used to fine-tune Qwen2.5-VL-7B-Instruct using QLoRA for tomato and mango disease diagnosis. The performance of the fine-tuned model for tomatoes was 93.7% while that of mangoes was 82.0%. The performances were higher than those of zero-shot learning which stood at 7.3% and 17.5% for tomatoes and mangoes, respectively. In terms of caption quality, the performance of the fine-tuned models was also better.

The results show that the proposed approach can improve both disease classification and the quality of generated diagnostic responses. At the same time, the per-class results show that some diseases remain difficult to distinguish, particularly Die Back and Anthracnose in mango. The evaluation was performed on held-out images from the respective datasets, so the reported results should not be taken as evidence of performance on uncontrolled field images.

Overall, PlantDx provides a structured and reproducible way of building knowledge-grounded supervision for agricultural VLMs. Its use across both tomato and mango shows that the same pipeline can be applied to different crop disease domains, providing a starting point for future work on additional crops and more challenging real-world agricultural images.

VII. FUTURE WORK

Future work will focus on extending PlantDx to additional crop and disease domains while retaining its knowledge-grounded and reproducible data-generation process. An important direction is the development of an image-grounded instruction dataset builder, which would allow the generated supervision to incorporate image-specific information while preserving the existing knowledge constraints. Further evaluation on field and mobile-phone images is also needed to assess how well the adapted models generalize beyond the controlled images used in the current experiments. In addition, the lower performance observed for challenging classes such as Die Back and Anthracnose motivates further investigation of disease categories with visually overlapping symptoms. Future experiments can also examine the individual contributions of the ontology, controlled vocabulary, concept models, caption templates, and knowledge-grounded supervision through ablation studies.

REFERENCES

[1] S. P. Mohanty, D. P. Hughes, and M. Salathé, “Using deep learning for image-based plant disease detection,” *Frontiers in Plant Science*, vol. 7, Art. no. 1419, 2016, doi: 10.3389/fpls.2016.01419.

[2] K. P. Ferentinos, “Deep learning models for plant disease detection and diagnosis,” *Computers and Electronics in Agriculture*, vol. 145, pp. 311–318, 2018, doi: 10.1016/j.compag.2018.01.009.

[3] E. C. Too, L. Yujian, S. Njuki, and L. Yingchun, “A comparative study of fine-tuning deep learning models for plant disease identification,” *Computers and Electronics in Agriculture*, vol. 161, pp. 272–279, 2019, doi: 10.1016/j.compag.2018.03.032.

[4] A. Kamilaris and F. X. Prenafeta-Boldú, “Deep learning in agriculture: A survey,” *Computers and Electronics in Agriculture*, vol. 147, pp. 70–90, 2018, doi: 10.1016/j.compag.2018.02.016.

[5] J. G. A. Barbedo, “Factors influencing the use of deep learning for plant disease recognition,” *Biosystems Engineering*, vol. 172, pp. 84–91, 2018, doi: 10.1016/j.biosystemseng.2018.05.013.

[6] A. Radford *et al.*, “Learning transferable visual models from natural language supervision,” in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, vol. 139, 2021, pp. 8748–8763.

[7] J. Li, D. Li, S. Savarese, and S. Hoi, “BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models,” in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, vol. 202, 2023, pp. 19730–19742.

[8] S. Bai *et al.*, “Qwen2.5-VL technical report,” *arXiv preprint* arXiv:2502.13923, 2025.

[9] E. J. Hu *et al.*, “LoRA: Low-rank adaptation of large language models,” in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2022.

[10] M. Gupta, A. Mangla, P. Desai, and R. Greer, “Self-consistency in vision-language models for precision agriculture: Multi-response consensus for crop disease management,” in *Proc. IEEE Int. Conf. Image Process. Workshops (ICIPW)*, 2025, pp. 185–190, doi: 10.1109/ICIPW68931.2025.11386079.

[11] P. Panchal, V. Polara, S. U, A. Baz, and S. K. Patel, “Deep learning-driven image captioning: Progress through transformers and large language models,” *PLOS ONE*, vol. 21, no. 3, Art. no. e0345012, 2026, doi: 10.1371/journal.pone.0345012.

[12] D. P. Hughes and M. Salathé, “An open access repository of images on plant health to enable the development of mobile disease diagnostics,” *arXiv preprint* arXiv:1511.08060, 2015.

[13] S. I. Ahmed *et al.*, “MangoLeafBD: A comprehensive image dataset to classify diseased and healthy mango leaves,” *Data in Brief*, vol. 47, Art. no. 108941, 2023, doi: 10.1016/j.dib.2023.108941.

[14] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “QLoRA: Efficient finetuning of quantized LLMs,” *arXiv preprint* arXiv:2305.14314, 2023.

[15] H. Liu, C. Li, Q. Wu, and Y. J. Lee, “Visual instruction tuning,” in *Advances in Neural Information Processing Systems*, vol. 36, pp. 34892–34916, 2023.

[16] W. Dai *et al.*, “InstructBLIP: Towards general-purpose vision-language models with instruction tuning,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[17] S. Xiong *et al.*, “Boosting crop disease recognition via automated image description generation and multimodal fusion,” *Computers and Electronics in Agriculture*, vol. 239, Art. no. 111082, 2025, doi: 10.1016/j.compag.2025.111082.

[18] Y. Wang *et al.*, “A large language model for multimodal identification of crop diseases and pests,” *Scientific Reports*, vol. 15, Art. no. 21959, 2025, doi: 10.1038/s41598-025-01908-0.

[19] Z. Bai *et al.*, “Hallucination of multimodal large language models: A survey,” *arXiv preprint* arXiv:2404.18930, 2024.

[20] H. Liu *et al.*, “A survey on hallucination in large vision-language models,” *arXiv preprint* arXiv:2402.00253, 2024.

[21] R. Yan *et al.*, “A knowledge graph for crop diseases and pests in China,” *Scientific Data*, vol. 12, Art. no. 222, 2025, doi: 10.1038/s41597-025-04492-0.

[22] H. Wu *et al.*, “Crop GraphRAG: Pest and disease knowledge base Q&A system for sustainable crop protection,” *Frontiers in Plant Science*, vol. 16, Art. no. 1696872, 2026, doi: 10.3389/fpls.2025.1696872. The article was published on January 5, 2026, despite 2025 appearing in the DOI.

[23] D. Hendrycks and K. Gimpel, “A baseline for detecting misclassified and out-of-distribution examples in neural networks,” in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2017.

[24] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “BLEU: A method for automatic evaluation of machine translation,” in *Proc. 40th Annual Meeting of the Association for Computational Linguistics*, pp. 311–318, 2002, doi: 10.3115/1073083.1073135.

[25] C.-Y. Lin, “ROUGE: A package for automatic evaluation of summaries,” in *Text Summarization Branches Out*, pp. 74–81, 2004.

[26] S. Banerjee and A. Lavie, “METEOR: An automatic metric for MT evaluation with improved correlation with human judgments,” in *Proc. ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization*, pp. 65–72, 2005.

[27] R. Vedantam, C. L. Zitnick, and D. Parikh, “CIDEr: Consensus-based image description evaluation,” in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4566–4575, 2015.

[28] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, “BERTScore: Evaluating text generation with BERT,” in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2020.

[29] P. Sahoo *et al.*, “A comprehensive survey of hallucination in large language, image, video and audio foundation models,” *arXiv preprint* arXiv:2405.09589, 2024.